

EX 6 – Demonstration of Apriori algorithm

AIM :

To demonstrate the Apriori algorithm on a transaction dataset to find frequent itemsets and generate association rules.

ALGORITHM :

1. Data Preparation

1. Import required libraries (Pandas, mlxtend)
2. Load the transaction dataset
3. Convert data into a suitable format (one-hot encoding / transactional format)

2. Apply Apriori Algorithm

1. Set minimum support threshold
 2. Generate frequent itemsets using the Apriori method
 3. Filter itemsets based on support value
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3. Generate Association Rules

1. Set minimum confidence threshold
2. Generate rules from frequent itemsets
3. Calculate metrics:
 - Support
 - Confidence
 - Lift

4. Analyze Results

1. Identify strong rules (high confidence & lift)
2. Interpret relationships between items
3. Present results in tabular/visual form

CODE :

```
import pandas as pd

from mlxtend.preprocessing import TransactionEncoder

from mlxtend.frequent_patterns import apriori, association_rules


dataset = [

    ['Milk','Bread','Butter'],

    ['Bread','Butter'],

    ['Milk','Bread'],

    ['Milk','Butter'],

    ['Bread','Butter'],

    ['Milk','Bread','Butter'],

    ['Milk','Bread'],

    ['Butter']

]


te = TransactionEncoder()

te_array = te.fit(dataset).transform(dataset)

df = pd.DataFrame(te_array, columns=te.columns_)


frequent_items = apriori(df, min_support=0.3, use_colnames=True)


rules = association_rules(frequent_items, metric="confidence", min_threshold=0.6)


print(frequent_items)

print(rules)
```

OUTPUT :

```
support      itemsets
0  0.750      (Bread)
1  0.750      (Butter)
2  0.625      (Milk)
3  0.500      (Bread, Butter)
4  0.500      (Bread, Milk)
5  0.375      (Milk, Butter)

antecedents  consequents  antecedent support  consequent support  support \
0  (Bread)    (Butter)          0.750          0.750    0.500
1  (Butter)   (Bread)          0.750          0.750    0.500
2  (Bread)    (Milk)           0.750          0.625    0.500
3  (Milk)     (Bread)          0.625          0.750    0.500
4  (Milk)     (Butter)         0.625          0.750    0.375

confidence   lift  representativity  leverage  conviction \
0  0.666667  0.888889        1.0 -0.06250    0.750
1  0.666667  0.888889        1.0 -0.06250    0.750
2  0.666667  1.066667        1.0  0.03125    1.125
3  0.800000  1.066667        1.0  0.03125    1.250
4  0.600000  0.800000        1.0 -0.09375    0.625

zhangs_metric  jaccard  certainty  kulczynski
0  -0.333333  0.500000  -0.333333  0.666667
1  -0.333333  0.500000  -0.333333  0.666667
2  0.250000  0.571429  0.111111  0.733333
3  0.166667  0.571429  0.200000  0.733333
4  -0.400000  0.375000  -0.600000  0.550000
```

RESULT :

Thus the above program is executed and the result is obtained.